Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education

PHYSICS **0625/42**

Paper 4 Extended Theory **October/November 2017**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2017 series for most Cambridge IGCSE®, Cambridge International A and AS Level components and some Cambridge O Level components.

® IGCSE is a registered trademark.

This syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **10** printed pages.

© UCLES 2017

www.root19.com +91 - 9969 353 391

Question	Answer	Marks
1(a)	$\rho = m / V$ in any form OR (m =) ρV OR (m =) $9000 \times 7.5 \times 10^{-5}$	**C1**
	(m =) 0.68 kg **accept** 680 g	**A1**
1(b)(i)	$W = m g$ in any form or (W =) $m g$ OR (W =) 0.68×10	**C1**
	(W =) 6.8 N	**A1**
1(b)(ii)	any **one** of: weight has direction / mass does not weight is a vector / mass is not weight varies / mass does not mass is amount of matter weight is a force / mass is not	**B1**
1(c)(i)	$\rho = h \rho g$ in any form OR (ρ =) $\rho / h g$ OR (ρ =) $560 / (0.027 \times 10)$	**C1**
	(ρ =) 2.1×10^3 kg / m^3	**A1**
1(c)(ii)	explains why there is a resultant downward force	**B1**

ROOT 19

Question	Answer	Marks
2(a)	accelerate / increase speed OR decelerate / decrease speed OR stop	**B1**
	change direction / move in a curve o.w.t.t.e.	**B1**
2(b)	change of shape OR size	**B1**
2(c)(i)	$F = m\,a$ in any form OR (a =) F / m OR (a =) 3500 / 1400	**C1**
	(a =) 2.5 m / s^2	**A1**
2(c)(ii)	$a = (v - u) / t$ in any form OR (t =) $(v - u) / a$ OR (t =) (30 – 0) / 2.5 OR 30 / 2.5	**C1**
	(t =) 12 s	**A1**
2(c)(iii)	friction / air resistance / drag	**B1**

ROOT 19

Question	Answer	Marks
3(a)	suitable fuel for a power station	**B1**
	any **three** from five: • thermal energy / heat (from fuel) • water / steam / gas heated OR steam produced • (steam / gas) turns / moves / drives turbine • (turbine) turns / moves / drives generator • 2 correct energy transfers	**B3**
3(b)	sun is energy source for plants / living matter (to grow) o.w.t.t.e.	**B1**
	plant / animal (remains compressed) into fuel OR carbon / chemical energy stored / trapped in plant / animal (remains)	**B1**
3(c)	not renewable (as fuel is consumed)	**M1**
	could only be replaced over very long time period (e.g. clearly > 50 years)	**A1**

ROOT 19

Question	Answer	Marks
4(a)(i)	any **one** of these six: • evaporation: at surface OR no bubbles form) pair 1 • boiling: throughout liquid OR bubbles form) • evaporation: at any temperature OR no heat needed) pair 2 • boiling: at specific temperature OR heat needed) • evaporation: affected by draught / surface area) pair 3 • boiling: not affected by draught / surface area)	**B1**
	any **one** pair of points	**B1**
4(a)(ii)	(it / rate) increases AND {more molecules have enough energy to escape OR break bonds}	**B1**
4(b)(i)	remains constant	**B1**
4(b)(ii)	$E = m\,l$ in any form OR (E =) $m\,l$	**C1**
	P = energy / t in any form OR (P =) energy / t	**C1**
	($P = 0.095 \times 2.3 \times 10^6 / (12 \times 60)$ =) 300 W	**A1**

Question	Answer	Marks
5(a)	any **three** of these five: • any sensible mention of the sun (as source of energy) • (thermal / heat / *IR* / electromagnetic) radiation • white (or clearly implied) surfaces absorb less or don't absorb • white (or clearly implied) surfaces reflect more • to keep house cooler OR to reduce thermal energy / heat transferred to house	**B3**
5(b)	decreases	**B1**

© UCLES 2017
www.root19.com

+91 - 9969 353 391

Question	Answer	Marks
6(a)(i)	diffraction	**B1**
6(a)(ii)	4 arcs between dashed lines centred vertically at centre of gap	**B1**
	any 3 wavelengths same as incident wavelengths including wavelength from wavefront in gap	**B1**
6(b)(i)	wavefronts have smaller angular width OR do not extend as far as dashed lines OR less (angular) spread	**B1**
6(b)(ii)	increased wavelength OR more spreading	**B1**
	use of $v=f\lambda$ OR increased wavelength	**B1**

Question	Answer	Marks
7(a)	real (answers in any order for **7(a)**)	**B1**
	enlarged OR magnified	**B1**
	Inverted OR upside down	**B1**
7(b)(i)	1st straight incident ray from close to point object to mirror correctly reflected, *i* = *r*	**M1**
	2nd straight incident ray <u>from point object</u> to mirror correctly reflected, *i* = *r*	**A1**
7(b)(ii)	BOTH reflected rays extended back to intersect behind mirror	**M1**
	BOTH reflected rays extended back <u>in straight lines</u> AND I in correct position AND {labelled OR clearly indicated}	**A1**

ROOT 19

www.root19.com +91 - 9969 353 391

Question	Answer	Marks
8(a)	$R_S = R_A + R_B$ in any form OR (R_S =) $R_A + R_B$ OR (R_S =) 4 + 8	**C1**
	(R_S =)12 (Ω)	**C1**
	(R_P =)1 / (1 / R_S + 1 / R_C) in any form OR (R_P =) $R_S R_C$ / (R_S + R_C) OR (R_P =) 1 / (1 / 12 + 1 / 6) OR (R_P =) (6 × 12) / 18	**C1**
	(R_P =) 4.0 Ω	**A1**
8(b)	V_8 = supply V × (8 / 12) OR = 24 × (8 / 12)	**C1**
	(V_8 =) 16 V	**A1**
	OR alternative route	
	I_8 = supply V / 12 OR = 24 / 12 OR = 2 (A)	**(C1)**
	(V_8 = 2 × 8 =) 16 V	**(A1)**

ROOT 19

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td rowspan="3">9(a)(i)</td><td>A (fixed)resistor
B thermistor
C L.E.D. OR light emitting diode</td><td></td></tr>
<tr><td>2 correct</td><td>B1</td></tr>
<tr><td>3 correct</td><td>B1</td></tr>
<tr><td>9(a)(ii)</td><td>any four from six:
• if cold / hot resistance of thermistor high / low
• if cold / hot voltage (across) thermistor high / low
• if cold / hot voltage of input to LED high / low
• if cold / hot there is current / no current in LED
• if cold LED lights / brighter
• if hot LED does not light / dimmer</td><td>B1</td></tr>
<tr><td rowspan="4">9(b)</td><td>
<table>
<tr><th>Row</th><th>A</th><th>B</th><th>C</th><th>(output of AND)</th><th>X</th></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>2</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>4</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>5</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>6</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>7</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>8</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table>
</td><td></td></tr>
<tr><td>row 1 of X correct – answer 0</td><td>B1</td></tr>
<tr><td>rows 2 AND 3 of X correct – both answers 0</td><td>B1</td></tr>
<tr><td>rows 4–8 of X correct – all answers 1</td><td>B1</td></tr>
</table>

© UCLES 2017
www.root19.com

+91 - 9969 353 391

Question	Answer	Marks
10(a)(i)	clockwise arrows on at least 3 circles	**B1**
10(a)(ii)	(magnetic) field becomes weaker / decreases (as distance from wire increases)	**B1**
10(b)(i)	any **four** from these six: • charge flows OR current in solenoid / wire / circuit • solenoid becomes magnet / magnetised • bolt becomes magnet / magnetised • (such that) unlike poles (of solenoid and bolt are) facing o.w.t.t.e. • bolt is attracted • bolt moves / (door) locks / spring stretched	**B4**
10(b)(ii)	solenoid OR bolt no longer magnetised OR bolt no longer attracted	**B1**
	(spring contracts and pulls) bolt back / bolt returns (to original position) / (door) unlocked	**B1**

ROOT 19

Question	Answer	Marks
11(a)	nucleon numbers balance each side of equation	**B1**
	proton numbers balance each side of equation	**B1**
	$^{0}_{-1}\beta$	**B1**
11(b)(i)	background radiation OR radiation from the environment	**B1**
	rocks / ground / buildings / food / space / weapons testing / nuclear accidents or waste / sun / air / radon / argon	**B1**
11(b)(ii)	random (variation)	**B1**
11(b)(iii)	clear evidence of subtracting 23 from (original) count	**C1**
	clear evidence of dividing original / corrected count by 4	**A1**
	clear evidence of adding 23 correctly to result after division	**A1**

ROOT 19